ECE 3308-002

Exploratory Data Analysis of NCAA Outcomes (2003-2025)

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Introduction

The objective of this analysis is to determine which engineered performance metrics most strongly predict winning in NCAA tournament play. Using game data from 2003–2025, four differential performance metrics were constructed based on the "Four Factors" of basketball analytics:

- Effective Field Goal Percentage (Diff_eFG)
- Turnover Rate (Diff_TOV)
- Offensive Rebound Rate (Diff_ORB)
- Free Throw Rate (Diff_FTR)

Each feature represents the difference between Team A and Team B within a single game. The target variable, Win_A, indicates whether Team A won (1) or lost (0). The dataset contains over 120,000 games and is balanced between wins and losses, ensuring that correlation results are not distorted by class imbalance. Teams were randomly assigned to Team A or Team B to prevent directional bias.

This analysis addresses three key questions:

1. Which of the four factors has the strongest statistical relationship with winning?
2. Do these variables visually separate winners from losers?
3. Do tournament seed patterns align with performance metrics, particularly in major upsets?

Dataset Overview

The dataset includes Division I men's basketball games from the 2003 through 2025 seasons. Regular season and tournament games were combined to create a comprehensive dataset. After filtering, the final dataset contains 120,264 games, with each row representing a single matchup.

The target variable Win_A indicates whether Team A won (1) or lost (0). The distribution is approximately balanced (≈49.9% wins and ≈50.1% losses), supporting unbiased statistical interpretation.

Four engineered differential features were constructed:

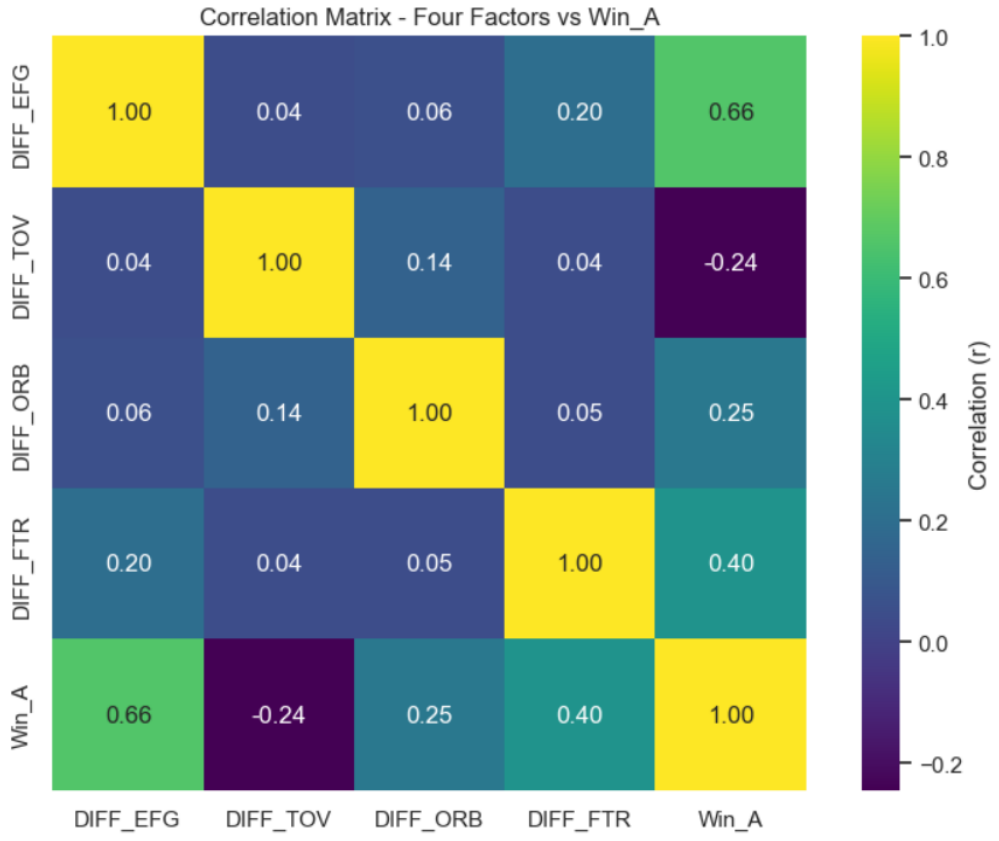
- Diff_eFG – Difference in effective field goal percentage
- Diff_TOV – Difference in turnover rate
- Diff_ORB – Difference in offensive rebound rate
- Diff_FTR – Difference in free throw rate

Each feature captures Team A's performance relative to Team B within the same game. Because these metrics are game-level differentials rather than season averages, they reflect the immediate statistical conditions that determined each outcome.

Descriptive statistics confirm that each differential feature has a mean near zero, validating correct random assignment of Team A and Team B.

Correlation Analysis

This correlation matrix was generated using the four differential features discussed previously and the target variable Win_A. These results show that Diff_eFG exhibits the strongest correlation with winning (r ≈ 0.66), which suggests that shooting efficiency is the most influential predictor among the four factors. Diff_TOV shows a pretty interesting relationship with winning as well, though its magnitude is notably lower than that of Diff_eFG. This shows that while ball security contributes to game outcome, it is less predictive than shooting efficiency. Diff_ORB and Diff_FTR demonstrate weaker correlations, suggesting more limited standalone predictive power. Overall the analysis provides clear statistical evidence that shooting efficiency (eFG) is more strongly associated with winning than turnover rate (TOV) within this dataset.



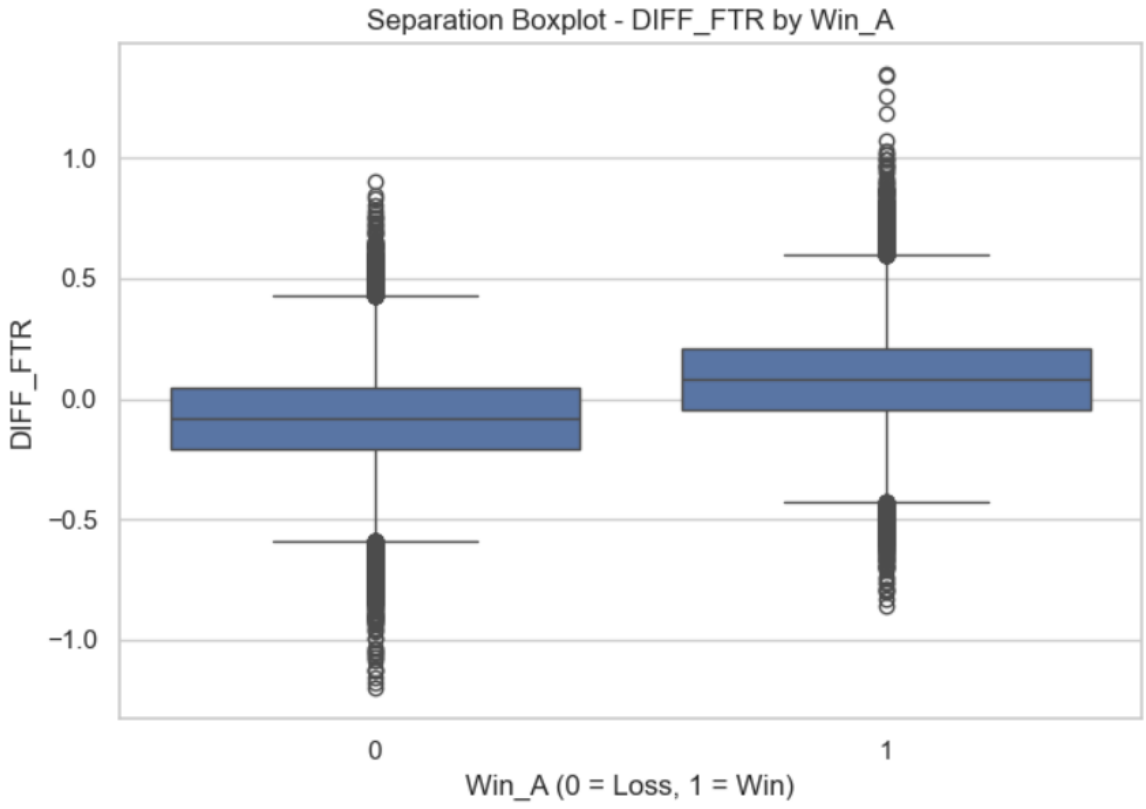
Feature Separation Analysis

To evaluate whether these features meaningfully distinguish winners from losers, boxplots were created for Diff_eFG and Diff_FTR, the two most strongly correlated variables.

The Diff_eFG boxplot shows a clear upward shift in the distribution for winning teams compared to losing teams. The median for winning games is consistently positive, while the median for losses is negative, with limited overlap between interquartile ranges. This strong visual separation supports the high correlation ($r \approx 0.66$) and confirms that shooting efficiency is a dominant driver of game outcomes.

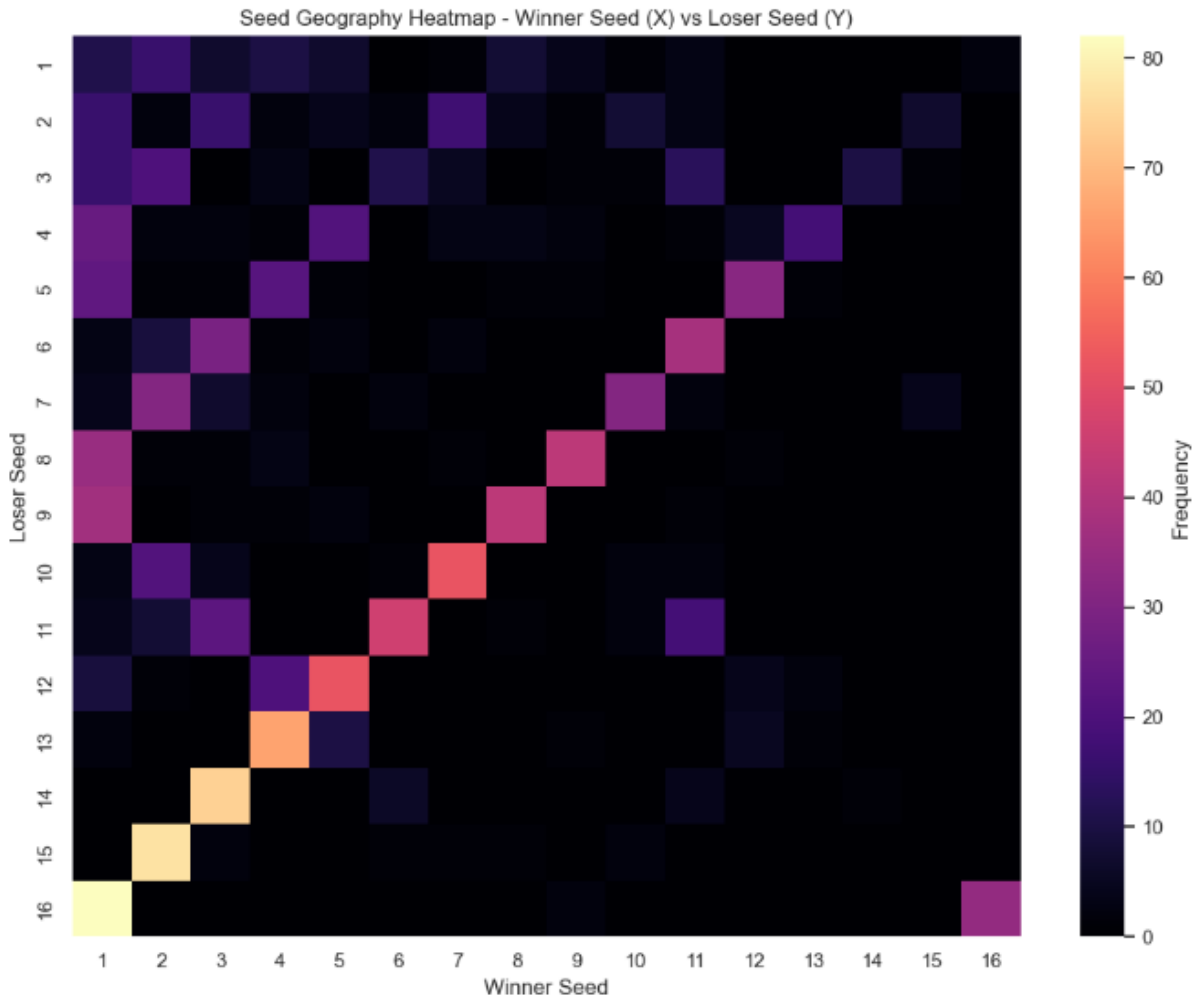
The Diff_FTR boxplot also demonstrates separation between winners and losers, though the overlap is greater. This aligns with its moderate correlation ($r \approx 0.40$), indicating that free throw rate contributes predictive value but is secondary to shooting efficiency.

Overall, the separation analysis reinforces the statistical findings from the correlation matrix.



Seed Geography Heatmap

The seed geography heatmap displays tournament game frequency by Winner Seed (x-axis) and Loser Seed (y-axis). A strong diagonal pattern is visible from (1,16) through (8,9), reflecting expected tournament structure where higher seeds consistently defeat lower seeds. The brightest region appears at (1,16), indicating the high frequency of 1-seeds defeating 16-seeds in early rounds. Cells above the diagonal represent upsets, where the winner has a numerically higher seed. These regions are significantly darker, confirming that such outcomes are rare.This visualization confirms that seeding strongly predicts tournament outcomes, though isolated upsets remain a defining characteristic of March Madness.



Upset Autopsy

The largest upset in the dataset occurred when a 16-seed defeated a 1-seed, producing a seed differential of 15. After adjusting the Four Factors to represent Winner minus Loser values, the underdog showed a positive Diff_eFG of approximately +0.038, indicating a slight shooting advantage.

However, the underdog performed worse in turnover rate, offensive rebounding, and free throw rate. This suggests that the upset was not driven by overwhelming statistical dominance across all factors. Instead, it appears that modest shooting efficiency gains combined with game-level variance contributed to the outcome.

Unlike typical tournament games where superior efficiency dominates, this upset reflects the inherent unpredictability of single-elimination play.

In other words, if a team shoots at an elite level for one night, even a top seed can fall. Sometimes it only takes a “Steph Curry” type performance to overturn expectations.

	123 <unnamed>
Season	2023.000000
Winner_Seed	16.000000
Loser_Seed	1.000000
Seed_Diff	15.000000
DIFF_EFG (Winner-Loser)	0.037888
DIFF_TOV (Winner-Loser)	-11.704389
DIFF_ORB (Winner-Loser)	-0.101961
DIFF_FTR (Winner-Loser)	-0.197200

Summary of Findings

After running the correlation analysis and reviewing the visualizations, it is clear that Diff_eFG is the strongest predictor of winning. With a correlation of approximately 0.66, shooting efficiency stands out by a wide margin compared to the other three factors. The boxplots reinforced this finding, showing a clear upward shift for winners and strong separation between winning and losing distributions. When teams shoot efficiently, they win.

Diff_FTR also demonstrated meaningful predictive value at approximately 0.40. Getting to the line and converting free throws contributes to success, especially in close tournament games where small margins matter. Diff_ORB and Diff_TOV were weaker individually but still relevant. Turnover differential was negatively correlated, which aligns with basketball fundamentals since giving away possessions reduces scoring opportunities. Offensive rebounding showed modest influence, supporting the idea that extra possessions still matter.

The 2023 upset of Fairleigh Dickinson over Purdue, a 16-seed defeating a 1-seed, illustrates that while seeding and efficiency metrics strongly predict outcomes, rare exceptions still occur. In that game, the underdog did not dominate across all four factors, reinforcing that single-elimination tournaments can produce unpredictable results even when statistical indicators favor the higher seed.

None of the four factors appear to be pure noise. Each captures a distinct dimension of performance: shooting efficiency, ball security, rebounding, and free throw generation. For the logistic regression model, I will retain all four features. However, effective field goal percentage is expected to carry the most weight.